Winding-free exact diagonalization of quantum spin ice: benchmarks and convergence criteria

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Exact diagonalization of quantum spin ice on a periodic cluster can generate noncontractible four-spin tunneling at second order, parametrically before the physical six-spin ring exchange. We formulate a nonperturbative winding-free remedy: exactly diagonalize a transported-dipole source family, canonically pull its isolated ice-descended band to a common basis, and project its zero Fourier component. At $(J_\pm, J_{\pm\pm})/J_{zz} = (0.046, 0)$, the character sequence converges from $M = 3$ to $M = 4$ within 0.50% in centered operator norm, with minimum ice overlap 0.762. The winding-free band moves the cubic-16 heat-capacity peak from $0.0255 J_{zz}$ to $0.00146 J_{zz}$, compared with the quantum Monte Carlo value $0.00110 J_{zz}$, while retaining the exact microscopic complement. The longitudinal response simultaneously moves from $0.08065 J_{zz}$ to a near-zero Γ line and $0.00535 J_{zz}$ at the three allowed X points. Thus the method is nonperturbatively converged on cubic-16, but the cluster is not yet a quantitative surrogate for bulk quantum spin ice.

Motivation.—Quantum spin ice (QSI) realizes a compact $U(1)$ gauge theory in a pyrochlore magnet: the two-in–two-out constraint is Gauss's law, defects are electric charges, and collective hexagon tunneling generates the emergent photon [1–3]. This low scale is experimentally consequential. In particular, the two heat-capacity features reported in $Ce_2Hf_2O_7$ motivate calculations that retain both the spin-ice crossover and quantum dynamics [4]. Numerically, the problem is divided by sign. Quantum Monte Carlo (QMC) reaches the zero-flux XXZ model and resolves its thermodynamics and dynamics [5, 6], whereas the opposite sign relevant to π-flux candidates has a sign problem. Exact diagonalization (ED) has no sign problem, but its periodic clusters alter the shortest allowed gauge processes.

We address that finite-size problem without replacing the microscopic model by an effective Hamiltonian. Perturbation theory is used only to identify the offending path and test its topology; production results come from exact microscopic eigenspaces. The controlled numerical limit is the transport-character resolution $M \to \infty$. On the sign-free side, winding removal must also improve convergence toward QMC. Exact removal of a known artifact is not by itself proof of convergence to the bulk.

The finite torus changes the leading process.—We study

$$H = J_{zz} \sum_{\langle ij \rangle} S_i^z S_j^z - J_\pm \sum_{\langle ij \rangle} (S_i^+ S_j^- + S_i^- S_j^+) + J_{\pm\pm} \sum_{\langle ij \rangle} (S_i^+ S_j^+ + S_i^- S_j^-), \qquad J_{zz} > 0. \tag{1}$$

For an S^z basis state $|s\rangle$, let $Q_t(s) = \sum_{i \in t} S_i^z(s)$. The ice-rule projector is simply

$$P_{\rm ice} = \sum_{\substack{s:\, Q_t(s)=0 \\ \text{for every } t}} |s\rangle\langle s|. \tag{2}$$

Thus $P_{\rm ice}$ is the orthogonal projector onto the subspace satisfying the ice rule on every tetrahedron. In the infinite pyrochlore lattice the shortest contractible process acting within this subspace is a hexagon. The protocol below includes both transverse couplings; the analytic scales quoted next and all present benchmark curves use the QMC-matched slice $J_{\pm\pm} = 0$. Its third-order matrix element has scale $g_6 = 12 |J_\pm|^3 / J_{zz}^2$ [1]. On a torus, however, a path may close through a periodic image. Cubic-16 contains 36 winding four-cycles in two winding classes [Figs. 1i.b) and 1i.c)] and 16 contractible hexagons [Fig. 1i.a)]. The four-spin process enters at second order with $g_4 = 4 J_\pm^2 / J_{zz}$, so $g_4/g_6 = J_{zz}/(3|J_\pm|)$ diverges in the nominally controlled limit. FCC-32 reduces but does not eliminate the problem: it has 24 winding four-cycles, 32 contractible hexagons, and 96 wrapping hexagons.

The distinction is visible before assigning any perturbative amplitude. Lift each exchange sequence γ to the periodic covering lattice and add its signed spin-transfer displacements. The two four-loops transport a finite dipole $\boldsymbol{q}_\gamma \neq 0$ across the torus, although they return to the same finite-cell spin configuration. By contrast, the alternating transfers on a contractible hexagon sum to $\boldsymbol{q}_\gamma = 0$. Our protocol projects the exact ice-connected band onto this winding-free ($\boldsymbol{q} = 0$) character, and therefore removes the former while retaining the latter.

Winding-free construction.—For an oriented bond, let $\boldsymbol{d}_{ij} = \boldsymbol{r}_j - \boldsymbol{r}_i - \boldsymbol{n}_{ij}^T L$ be its displacement and $\boldsymbol{c}_{ij} = \boldsymbol{r}_i + \boldsymbol{r}_j - \boldsymbol{n}_{ij}^T L$ its pair-flip coordinate in the periodic covering lattice. We diagonalize the explicitly sourced microscopic Hamiltonian

$$H(\boldsymbol{\theta}) = H_0 - J_\pm \sum_{\langle ij \rangle} \left(e^{2i\boldsymbol{\theta}\cdot\boldsymbol{d}_{ij}} S_i^+ S_j^- + e^{-2i\boldsymbol{\theta}\cdot\boldsymbol{d}_{ij}} S_i^- S_j^+ \right) + J_{\pm\pm} \sum_{\langle ij \rangle} \left(e^{-2i\boldsymbol{\theta}\cdot\boldsymbol{c}_{ij}} S_i^+ S_j^+ + e^{2i\boldsymbol{\theta}\cdot\boldsymbol{c}_{ij}} S_i^- S_j^- \right). \tag{3}$$

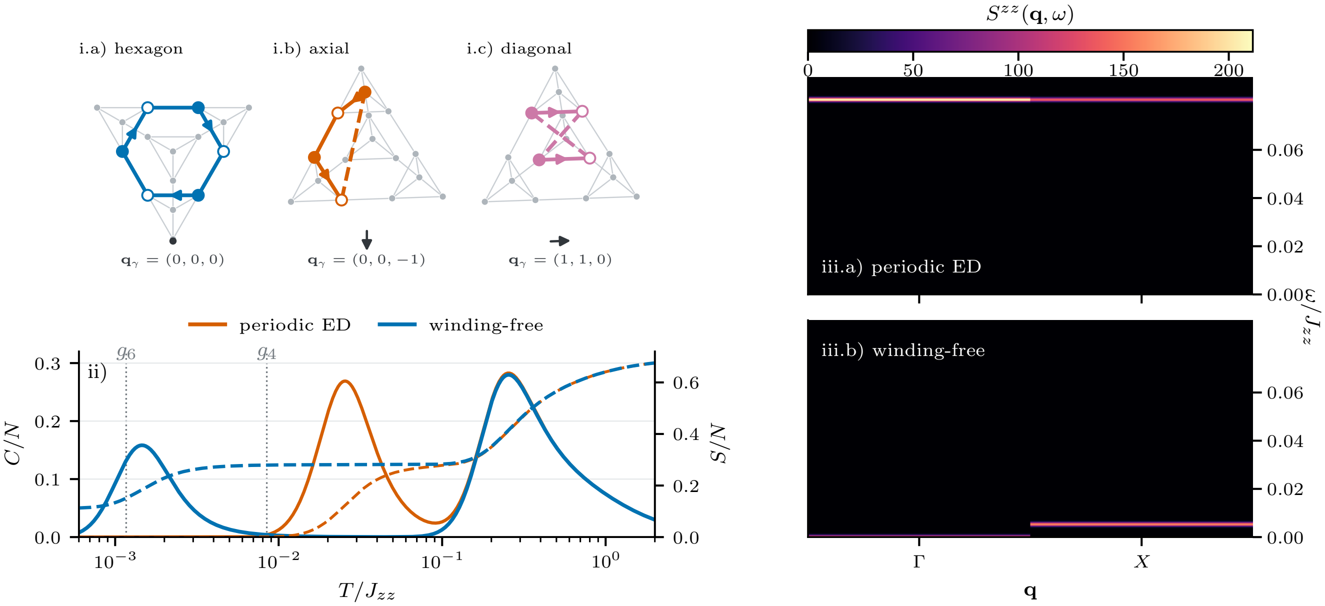

FIG. 1. Finite-torus transport and its winding-free removal on cubic-16. i.a) The contractible hexagon, viewed normal to its plane, has exact D_6 geometry and $\boldsymbol{q}_\gamma = 0$. i.b), i.c) Axial and diagonal winding four-loops close through periodic bonds (dashed) and carry $\boldsymbol{q}_\gamma \neq 0$. Gray sites and bonds show the hosting pyrochlore cluster; colored arrows are spin transfers and the dark marker or arrow is their resultant $\boldsymbol{q}_\gamma = 2\sum_\ell \Delta\boldsymbol{p}_\ell$. ii) Specific heat (solid, left axis) and entropy (dashed, right axis) at $(J_\pm, J_{\pm\pm})/J_{zz} = (0.046, 0)$; vertical lines mark g_6 and g_4. iii.a), iii.b) Periodic and winding-free longitudinal response on common linear scales, with X averaged over its three symmetry-equivalent representatives. The same canonically dressed microscopic S^z probe is used in both panels.

Equation (3) is the full microscopic finite-cluster Hamiltonian, not an effective Hamiltonian. No expansion or virtual-state denominator enters its definition. For a sequence γ of microscopic moves, each step ℓ acts on the stored oriented bond (i_ℓ, j_ℓ). Its lifted polarization increment is

$$\Delta\boldsymbol{p}_\ell = \begin{cases} -\boldsymbol{d}_{ij}, & T_\ell = S_i^+ S_j^-, \\ +\boldsymbol{d}_{ij}, & T_\ell = S_i^- S_j^+, \\ +\boldsymbol{c}_{ij}, & T_\ell = S_i^+ S_j^+, \\ -\boldsymbol{c}_{ij}, & T_\ell = S_i^- S_j^-. \end{cases} \tag{4}$$

Multiplication of the source phases gives $e^{-i\boldsymbol{\theta}\cdot\boldsymbol{q}_\gamma}$ with $\boldsymbol{q}_\gamma = 2\sum_{\ell\in\gamma}\Delta\boldsymbol{p}_\ell$. Completed ice-to-ice paths have integer $\boldsymbol{q}_\gamma$: contractible paths have $\boldsymbol{q}_\gamma = 0$, whereas the targeted paths with nonzero net winding transport have $\boldsymbol{q}_\gamma \neq 0$. This phase-product statement is an exact identity between microscopic matrix elements.

The symbol $P_{\boldsymbol{\theta}}$ denotes the exact spectral projector of $H(\boldsymbol{\theta})$ onto the isolated band continuously connected to the ice-rule subspace as the transverse couplings are turned on,

$$P_{\boldsymbol{\theta}} = \frac{1}{2\pi i}\oint_{\Gamma_{\boldsymbol{\theta}}} (z - H(\boldsymbol{\theta}))^{-1} dz, \qquad \operatorname{rank} P_{\boldsymbol{\theta}} = \operatorname{rank} P_{\text{ice}}. \tag{5}$$

It is distinct from the fixed ice projector. The polar map

$$W_{\boldsymbol{\theta}} = P_{\boldsymbol{\theta}} P_{\text{ice}} (P_{\text{ice}} P_{\boldsymbol{\theta}} P_{\text{ice}})^{-1/2} \tag{6}$$

canonically identifies the exact band with the fixed ice-rule reference subspace. The pulled-back operator is $h(\boldsymbol{\theta}) = W_{\boldsymbol{\theta}}^\dagger H(\boldsymbol{\theta}) W_{\boldsymbol{\theta}}$. We define

$$h_M = \frac{1}{M^3} \sum_{\boldsymbol{m}\in\mathbb{Z}_M^3} h(2\pi\boldsymbol{m}/M). \tag{7}$$

Equation (7) retains $\boldsymbol{q} = 0 \pmod M$ and converges to the continuous zero-character integral as $M \to \infty$. A closing band gap or a singular $P_{\text{ice}} P_{\boldsymbol{\theta}} P_{\text{ice}}$ is an explicit failure of the method. The construction is nonperturbative on the finite cluster because $H(\boldsymbol{\theta})$, $P_{\boldsymbol{\theta}}$, and the inverse square root in Eq. (6) are evaluated at the physical couplings, without a $J_\pm/J_{zz}$ or $J_{\pm\pm}/J_{zz}$ expansion. A formal expansion would recover all mixed orders in the two transverse couplings; the production calculation instead evaluates this function directly and projects its zero Fourier coefficient. Low-order Schrieffer–Wolff rows [7] only audit the loop interpretation. For thermal observables we define the hybrid replacement spectrum

$$Z_{\text{wf}} = Z_{\text{full}} - Z_{\text{periodic band}} + Z_{\text{wf band}}, \tag{8}$$

and apply the same identity to the first two energy moments. It leaves the microscopic complement intact, but it is a diagnostic spectrum rather than the spectrum of the original $H(0)$. Derivations and numerical details are given in the Supplemental Material [8].

Internal tests.—Direct row enumeration through third order gives a binary topology test. On both clusters, Eq. (7) removes 100% of winding four-loop weight and wrapping-hexagon weight and retains 100% of contractible-hexagon weight. The row classification and retained weight are defined explicitly in the Supplemental Material [8]. This is an order-two/order-three audit of the transport label; the exact-band projection itself is not truncated at either order. At $(J_\pm, J_{\pm\pm})/J_{zz} = (0.046, 0)$, cubic-16 moves from the periodic $T_\mathrm{p}/J_{zz} = 0.02554$ to the winding-free value 0.001252. FCC-32 gives 0.008770 and 0.000608 respectively within the same third-order diagnostic. These FCC-32 values are not production thermodynamics.

The exact cubic-16 calculation uses one source convention at every M. The zero-source band has minimum ice overlap 0.762 and fixed-S^z gap $0.627J_{zz}$. All $M = 3, 4$ source points have larger overlaps and gaps, with maximum Ritz residual 3.7×10^{-12}. The centered operator changes by 6.78% from $M = 2$ to 3, but only 0.498% from $M = 3$ to 4, passing the 5% character gate. The low-order audit shows that $M = 2$ already removes the shortest winding four-loops while retaining every contractible hexagon [the retained-weight table in the Supplemental Material [8]]. It is not an all-order zero-transport projector: every nonzero $\boldsymbol{q} = 2\boldsymbol{k}$ is still aliased with zero. The sizable $M = 2$ to 3 change and subsequent $M = 3$ to 4 agreement therefore test higher-order characters, not additional removal of the audited shortest loops.

Winding-free dynamics.—We hold the physical probe fixed while changing the band Hamiltonian. Let W_0 be the exact zero-source isometry in Eq. (6) and define

$$\widetilde{S}^z_a(\boldsymbol{q}) = W_0^\dagger \left(\sum_{j\in a} e^{-2\pi i \boldsymbol{q}\cdot\boldsymbol{r}_j} S^z_j \right) W_0. \tag{9}$$

For either $h(\mathbf{0})$ or $h_{M=4}$, we compute

$$S^{zz}(\boldsymbol{q},\omega) = \frac{1}{N d_0} \sum_{g\in\mathcal{G}} \sum_{n\notin\mathcal{G}} \sum_{a=0}^{3} |\langle n|\widetilde{S}^z_a(\boldsymbol{q})|g\rangle|^2 \times G_\eta[\omega - (E_n - E_g)], \tag{10}$$

where $\mathcal{G}$ is the ground multiplet and $\eta = 6 \times 10^{-4} J_{zz}$. Cubic-16 has only four translation momenta, Γ and three symmetry-related X points, which reduce to the two classes shown below.

The zero-source canonical isometry reproduces the cached periodic operator to 2.7×10^{-14} relative error. The periodic ground state is unique; the winding-free ground multiplet has dimension three. The X-point centroid moves from $0.08069J_{zz}$ to $0.005372J_{zz}$, demonstrating that the construction reorganizes both energies and matrix elements rather than merely shifting a thermodynamic peak.

External QMC gate.—Huang *et al.* simulate precisely Eq. (1) at $(J_\pm, J_{\pm\pm})/J_{zz} = (0.046, 0)$ using continuous-time worm QMC and report thermodynamics together with QMC-SAC dynamics on 8^3 primitive cells [6]. A vector extraction of their curves gives low- and high-temperature peaks at $T/J_{zz} = 0.001100$ and 0.2105. The converged $M = 4$ result places the low peak at $0.001463J_{zz}$ and reduces the low-temperature heat-capacity RMSE from 0.0859 to 0.02327, a 72.9% improvement [Fig. 1ii)]. It nevertheless misses the preregistered 0.02 tolerance. Entropy also fails: its RMSE is 0.04509, again above 0.02 [Fig. 1ii)]. The same paper provides the next dynamics gate, S^{zz} and S^{+-} at $T/J_{zz} = 0.001$, 0.04, and 0.1; matching its momentum points, normalization, and SAC resolution remains to be done.

Outlook.—The exact $M = 3, 4$ agreement establishes winding removal beyond low-order perturbation theory, but cubic-16 remains outside the QMC heat and entropy tolerances. The remaining campaign is a character-converged FCC-32 ABC/XYZ calculation followed by a $Ce_2Hf_2O_7$ heat-capacity fit; entropy and matched neutron channels are holdout tests, not fit normalizations.

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[8] (2026), see Supplemental Material for the canonical block construction, transported-dipole projection, exact-band limit, thermodynamic replacement identities, geometry, and benchmark provenance.